Mid-Progress Report: Equipment Predictive Maintenance and Fault Diagnosis System[EPMFDS] using Classical Machine Learning Techniques

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Abstract

This project explores classical machine learning techniques for predictive maintenance using the NASA C-MAPSS FD001 dataset. Five models have been implemented and evaluated: Nearest-Neighbors, Support Vector Machine, Decision Tree, Linear Regression, and Bayesian Learning using RMSE and MAE metrics. This report presents the methodology, experimental results, and future directions.
Keywords: Predictive Maintenance, Classical Machine Learning, Equipment Failure, Regression

Contents

1 Introduction

Predictive maintenance aims to anticipate equipment failures and minimize downtime. This project applies classical machine learning methods to forecast the Remaining Useful Life (RUL) of machinery components using sensor data. The focus is on methodical model evaluation and optimization.

2 Approaches Tried

2.1 K-Nearest Neighbors (KNN)

Predicts RULE based on similar past sensor readings. Serves as a baseline model.

2.2 Support Vector Machine (SVM)

Learns decision boundaries in the feature space for predictive maintenance.

2.3 Decision Tree

Uses hierarchical decision-making based on sensor values for better interpretability.

2.4 Linear Regression

Establishes linear relationships between sensor readings and RUL.

2.5 Bayesian Learning

Applies probabilistic reasoning to account for uncertainty in predictions.

3 Experiments and Results

Models were trained and tested in the C-MAPSS FD001 dataset. Performance is compared using RMSE and MAE metrics.

Model	Training RMSE	Test RMSE
K-Nearest Neighbors	XX.X	XX.X
Support Vector Machine	XX.X	XX.X
Decision Tree	XX.X	XX.X
Linear Regression	XX.X	XX.X
Bayesian Learning	XX.X	XX.X

Table 1: Performance Metrics of Models

4 Summary

The project has successfully implemented classical machine learning models for predictive maintenance. Future steps include hyperparameter tuning and model fusion techniques.

A Contribution of Each Member

1. **Aditya Kashyap:** Work on implementation of the SVM model, managed GitHub repository and preprocessing.
2. **Asadullah Faisal:** Work on the Bayesian learning model, researched dataset.
3. **Sunny Meena:** Work on KNN model, assisted in data preprocessing.
4. **Faiz Imdad:** Work on the decision tree model, contributed to the visualizations.
5. **Vishal Jharwal:** Work on Linear Regression model, assisted in performance evaluations.